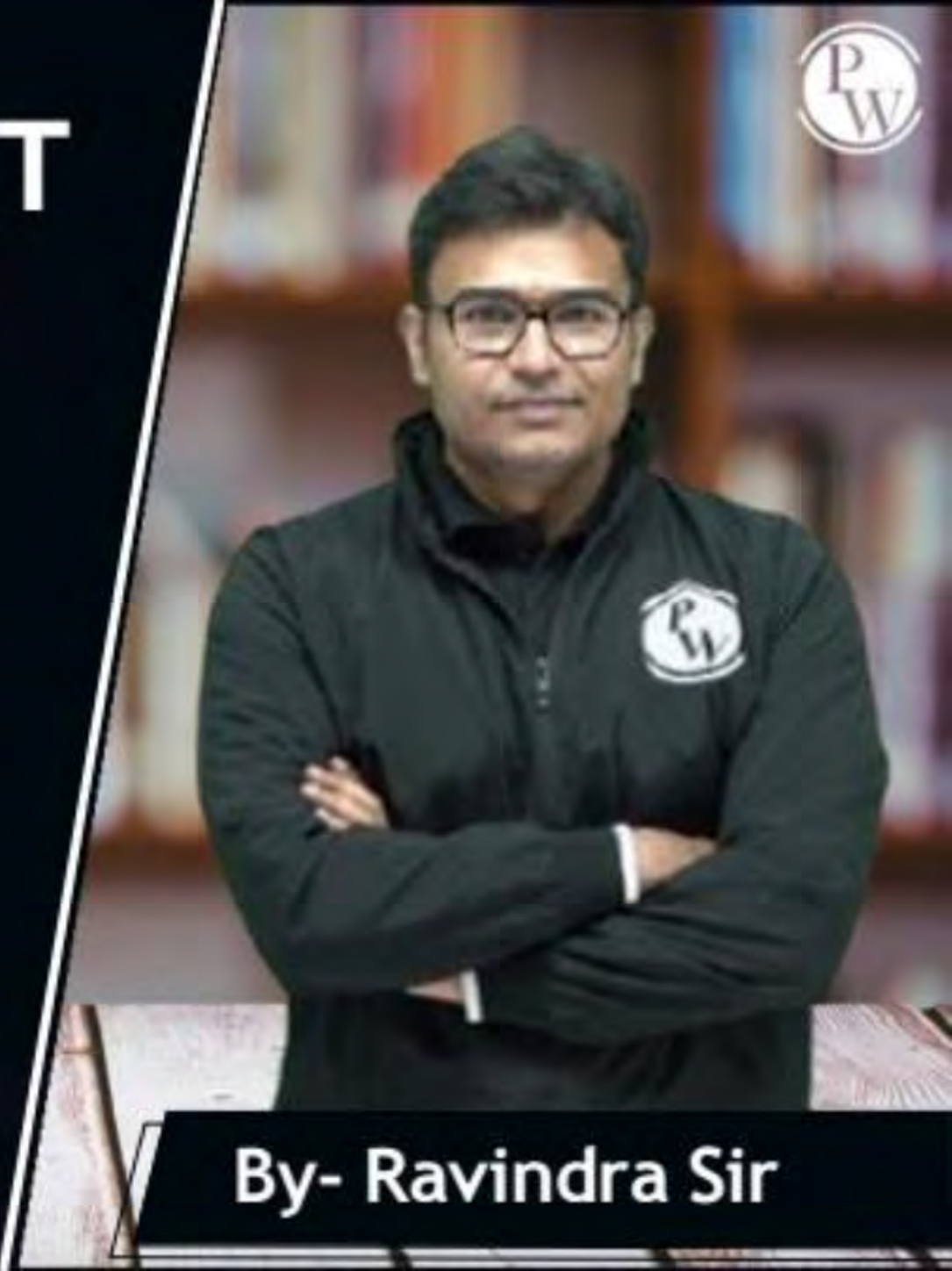
Computer Science & IT

COMPLIER DESIGN
(CD)

Parsers

Lecture No. 12



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Pauses

Topics to be Covered



- Topic
- Topic
- Topic

Parvan

Join Telegram

@RBRSIRPW





Inspiring Stories : Johanna Eck

Johanna Eck – The German Widow Who Resisted in Silence

Background: A Berlin housewife and former nurse during WWII's Nazi era.

Struggles: Risked everything to help Jewish friends when everyone else looked away.

Achievements: Hide four people in her apartment at great risk and was later honored as one of the Righteous Among the Nations.

Impact: She saved lives when it could have cost her own. If she could protect hope in danger, you can protect focus in chaos.



Inspiring Stories : Sasi Sekar Krish

Sasi Sekar Krish

Background: A VLSI researcher who lost his drive to start a business due to parental pressure to study more and take a corporate job.

Education: PhD from IISc Bangalore in VLSI algorithms.

Career Achievements: Left a stable job at Wipro to start NanoPix, making machines to help cashew farmers grade produce, and was the only one on his team willing to take the risk.

Impact: Used niche engineering skills to help small farmers earn more and reach better markets.



Inspiring Stories : Elizabeth Storie

Elizabeth Storie – The Seamstress Who Spoke Truth to Power

Background: A 19th-century hat maker from Glasgow's working class.

Struggles: Poisoned and disfigured by medical malpractice, and ignored by courts and officials.

Achievements: Published a book telling her story with documents to show the injustice.

Impact: She used silence as a tool to win justice. If she could fight unfairness with writing and proof, you can fight confusion with code and clarity.



Inspiring Stories : Rahul Panicker

Rahul Panicker

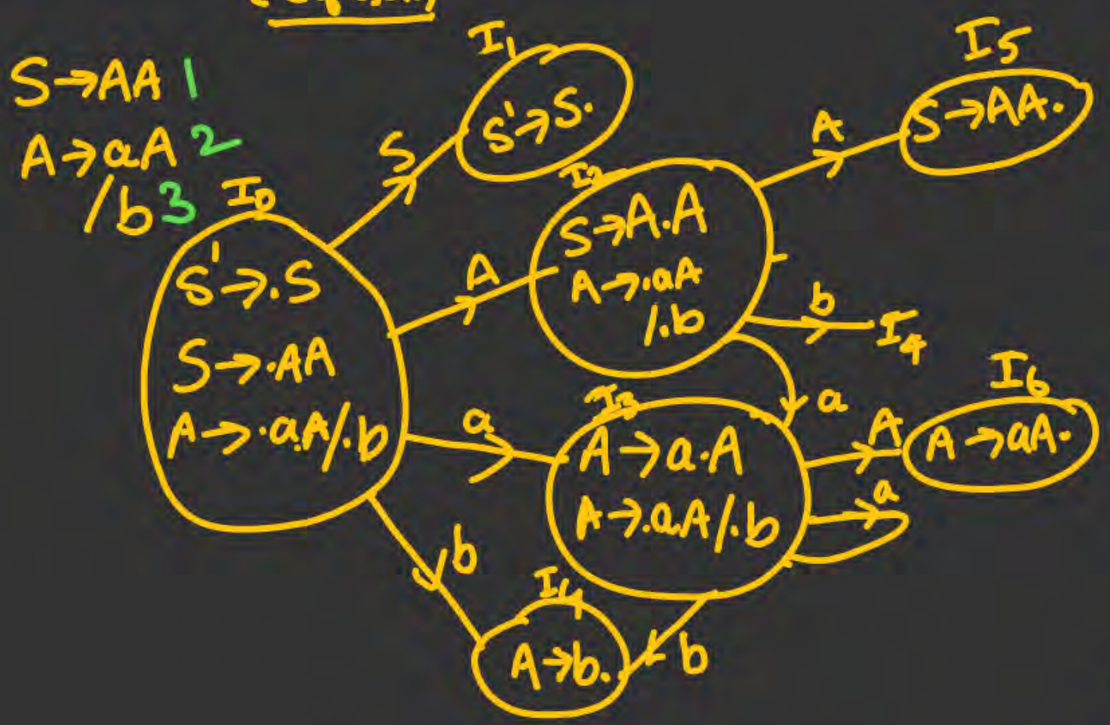
Background: From Kerala, he used technical skills and care for people to fight healthcare unfairness.

Education: Got MS and PhD from Stanford (after IIT Madras), but his social work stays true to his roots and goals.

Career Achievements: Co-founded Embrace Innovations, which made a cheap infant warmer to help premature babies.

Impact: Thousands of newborns in India and 15+ countries have benefited, greatly improving newborn care.

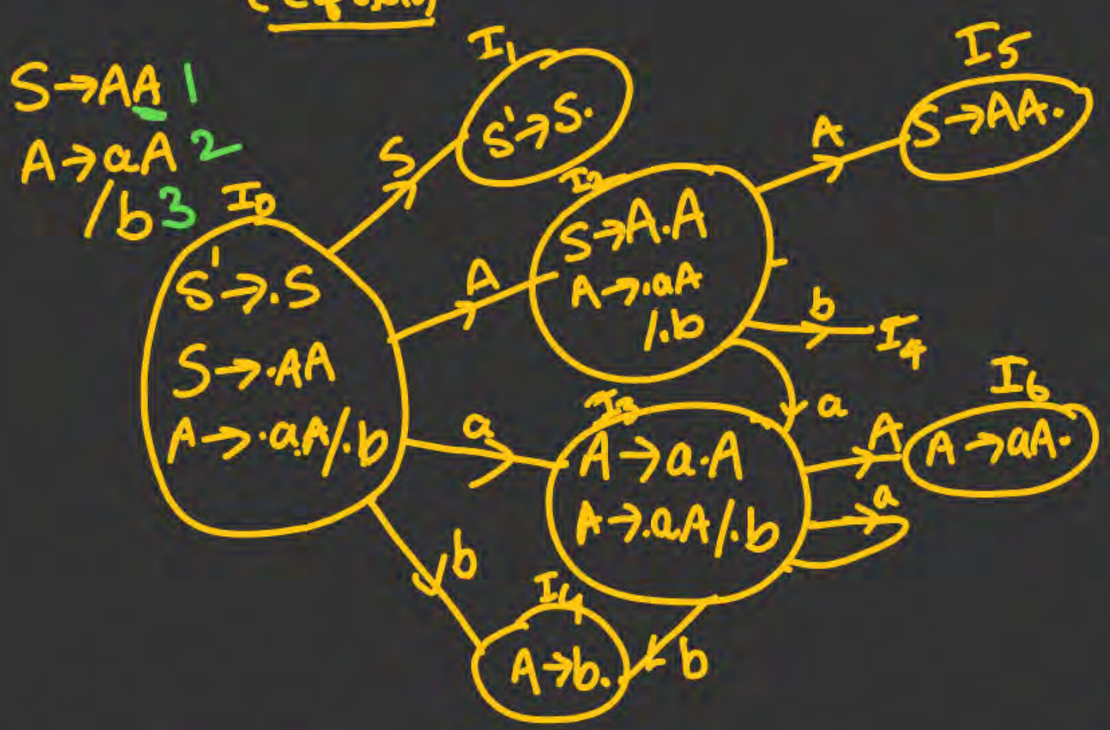
CC of LR(0)



LR(0) parsing table

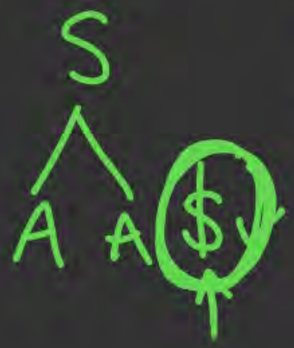
	action			Goto	
	a	b	$	A	S
0	S_3	S_4		2	1
1			acc		
2	S_3	S_4		5	
3	S_3	S_4		6	
4	γ_3	γ_3	γ_3		
5	γ_1	γ_1	γ_1		
6	γ_2	γ_2	γ_2		

CC of LR(0)



follow(S) = {$}

follow(A) = {a, b, $}



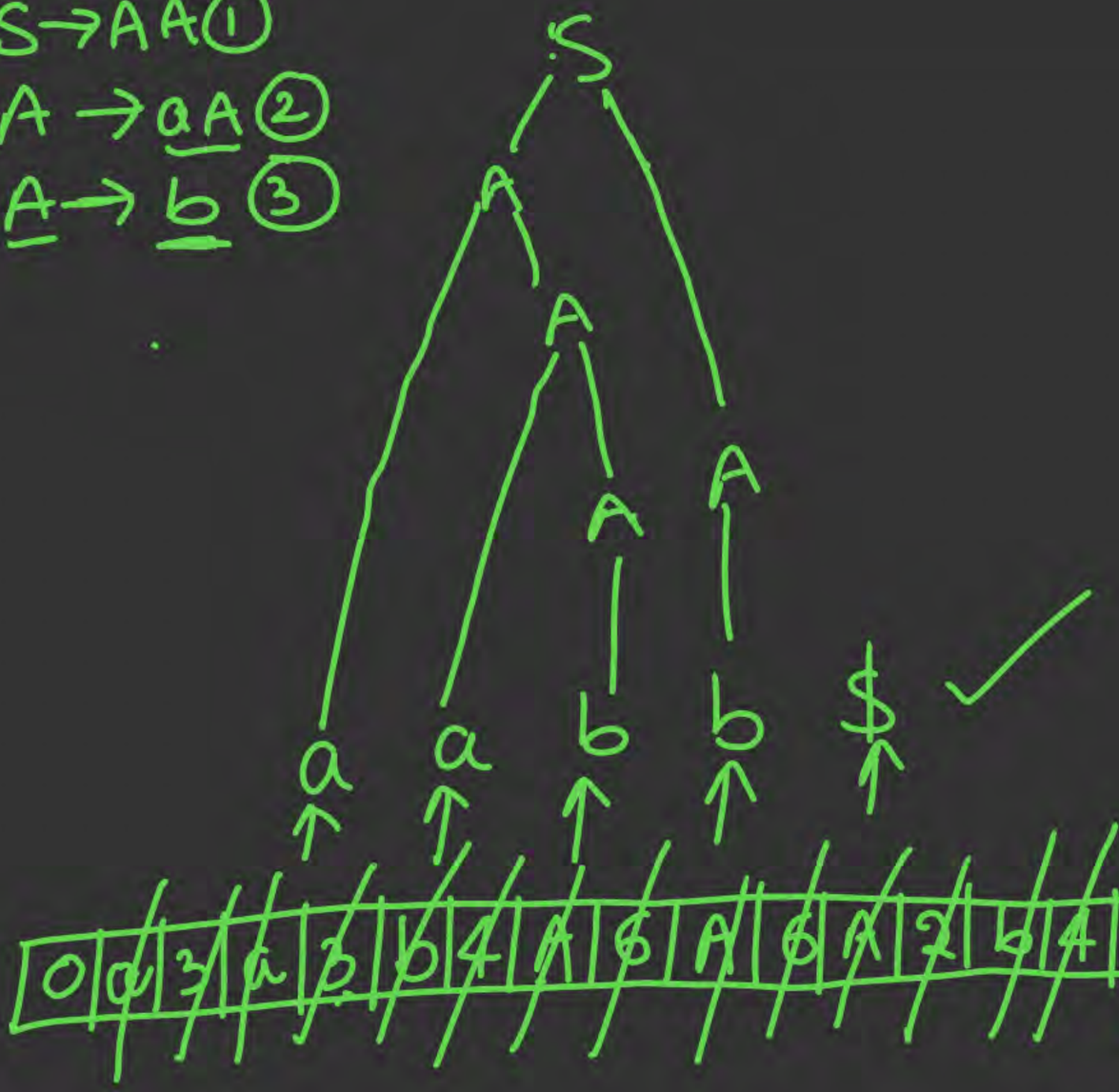
SLR(1) parsing table

	action			Goto	
	a	b	$	A	S
0	S_3	S_4		2	1
1			acc		
2	S_3	S_4		5	
3	S_3	S_4		6	
4	r_3	r_3	r_3		
5			r_1		
6	r_2	r_2	r_2		

$S \rightarrow AA$ (1)

$A \rightarrow aA$ (2)

$A \rightarrow b$ (3)



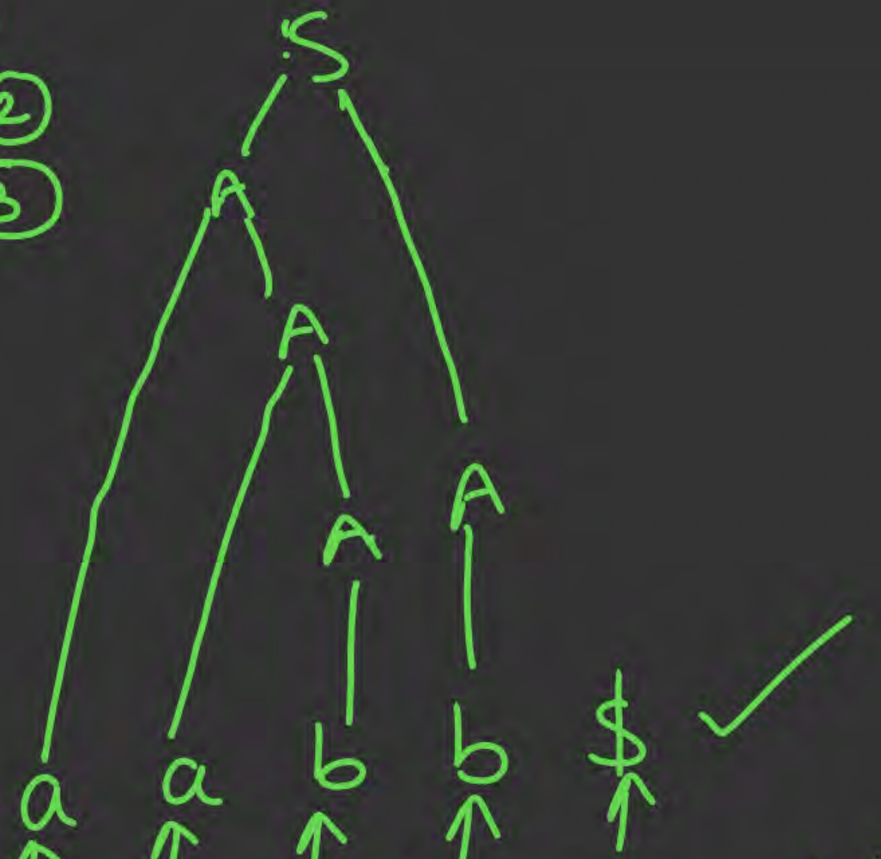
LR(0) parsing table

	action			Goto	
	a	b	$	A	S
0	S_3	S_4		2	1
1			acc		
2	S_3	S_4		5	
3	S_3	S_4		6	
4	γ_3	γ_3	γ_3		
5	γ_1	γ_1	γ_1		
6	γ_2	γ_2	γ_2		

$S \rightarrow AA$ (1)

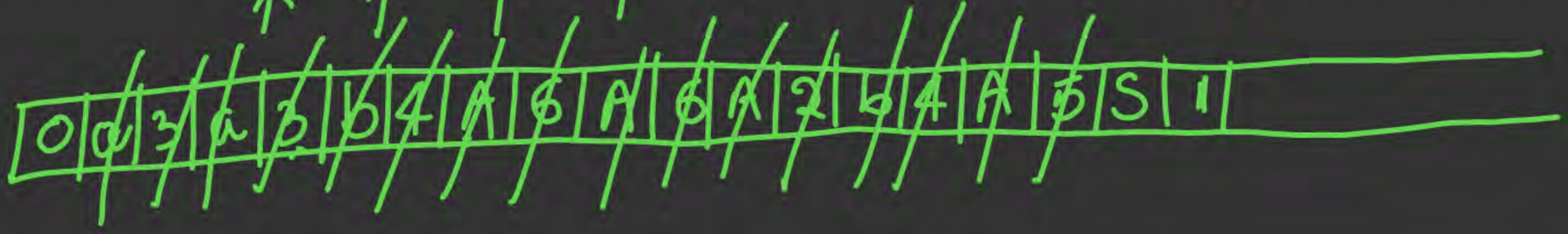
$A \rightarrow aA$ (2)

$A \rightarrow b$ (3)



LR(0) parsing table

	action			Goto	
	a	b	$	A	S
0	S_3	S_4		2	1
1			acc		
2	S_3	S_4		5	
3	S_3	S_4		6	
4					
5					
6					



SLR(1) parsing table :-

In case of Shift moves and goto moves LR(0) and SLR(1) are same but only difference is reduce action. Reduce actions are placed in entire row in LR(0) parsers. In SLR(1) parsers, the reduce move will be only in the follow of LHS. We are using 1 look ahead That is why this is called SLR(1). '1' means '1' lookahead. SLR(1) is more powerful than LR(0) because the reduce moves in SLR(1) are less. Blank spaces are more. So error entries are more. So SLR(1) detects more errors than LR(0).

Advantages of LR(0) parser

→ It is very simple to construct LR(0) parser compared to other LR parsers

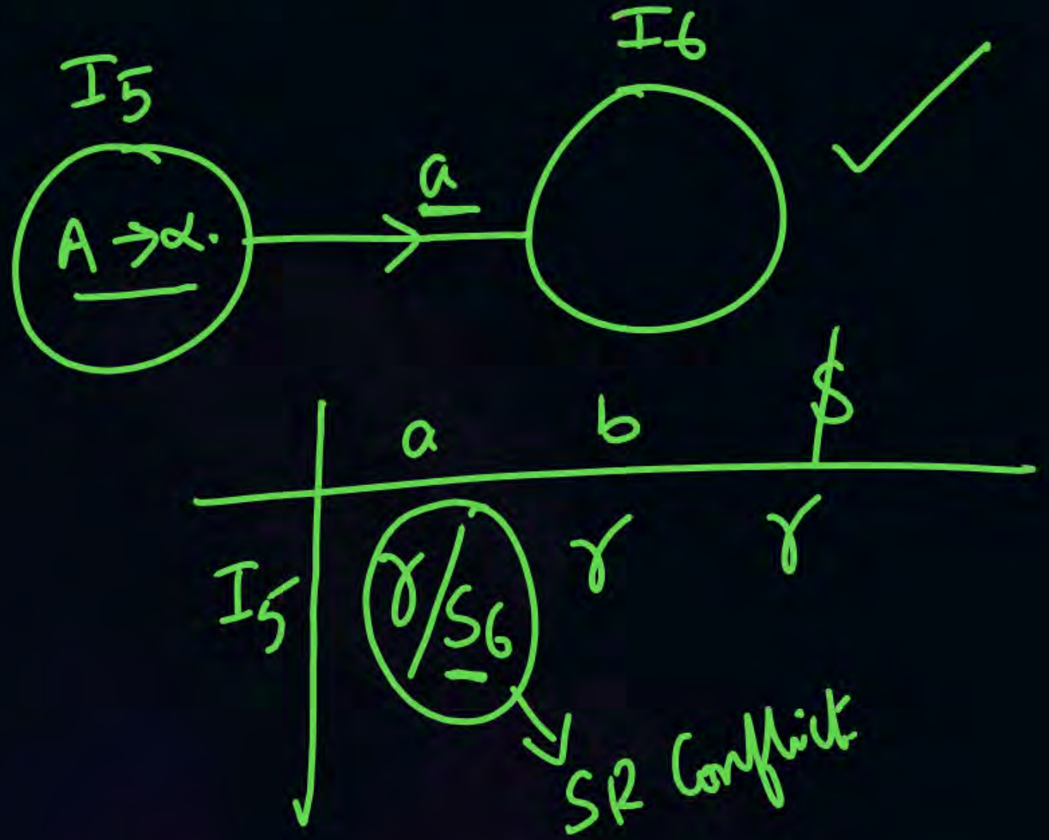
→ Each row in the table defines unique action, that is, either shift or reduce

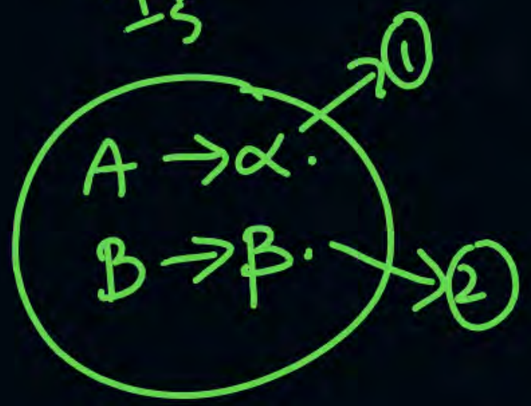
Disadvantages of LR(0) parser:-

→ The LR(0) parser can be used to parse only a small class of grammars.

→ look ahead is not used in making parsing decisions.

Testing whether a grammar is LR(0) :-

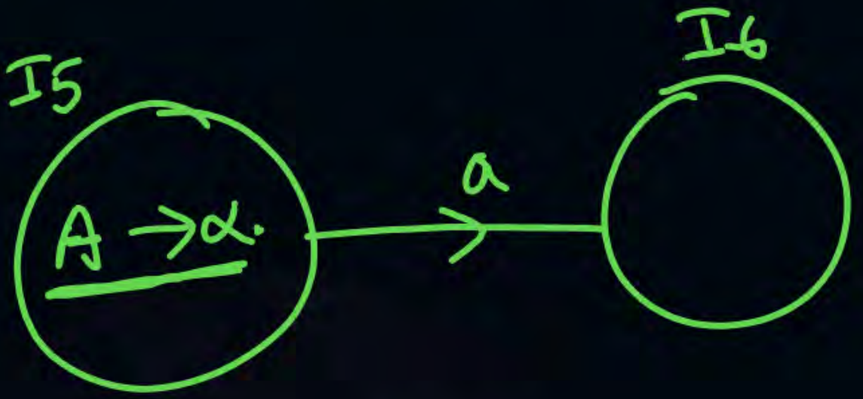




	a	b	$
5	r_1/r_2	r_1/r_2	r_1/r_2

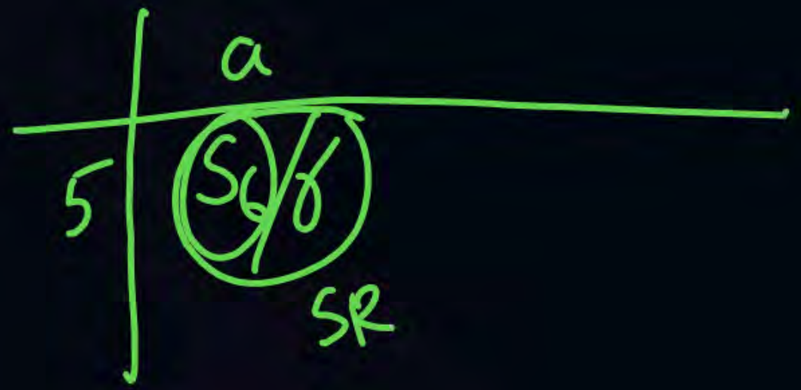
rr Conflict

Testing whether a grammar is SLR(1) :-

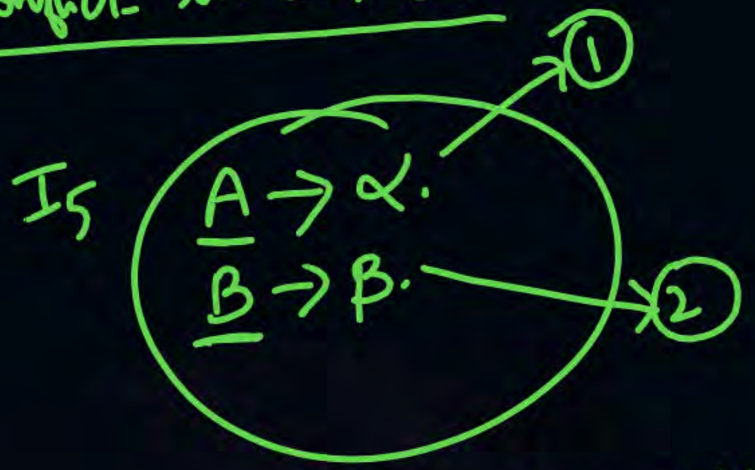


if follow(A) has 'a'

SR conflict



RR Conflict in SLR(1) :-



follow (A) = {a, c}
follow (B) = {a, b}

Foll(A) ∩ Foll(B) ≠ ∅

	a	b	c
5	r_1/r_2		

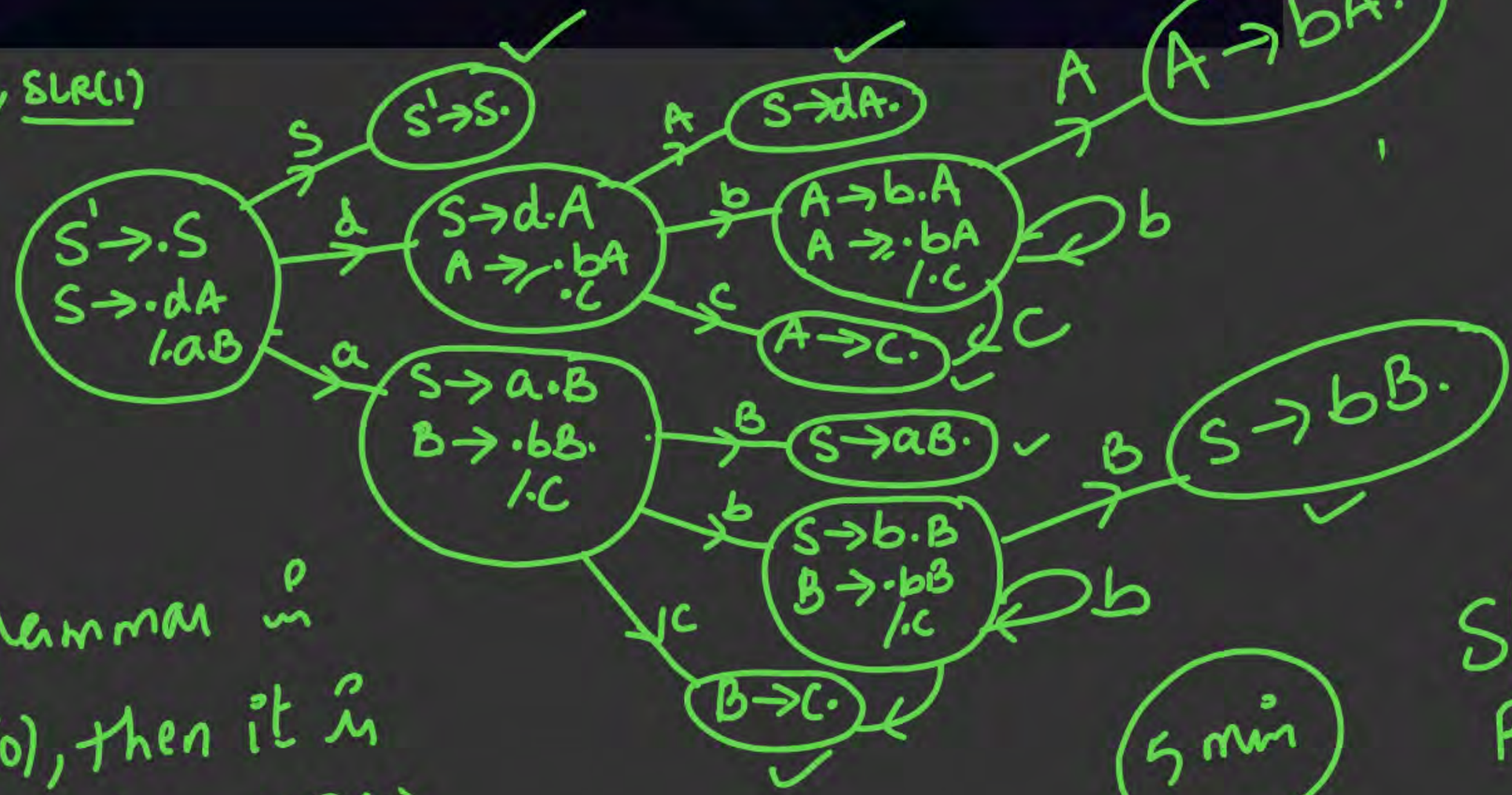
LR(0) → SLR(1)

LL(1), LR(0), SLR(1)

$S \to dA \mid aB$

$A \to bA \mid c$

$B \to bB \mid c$



If a grammar is LR(0), then it is always SLR(1)

5 min

S R
R R

LL(1) LR(0) SLR(1)

$S \to A / a$

$A \to a$

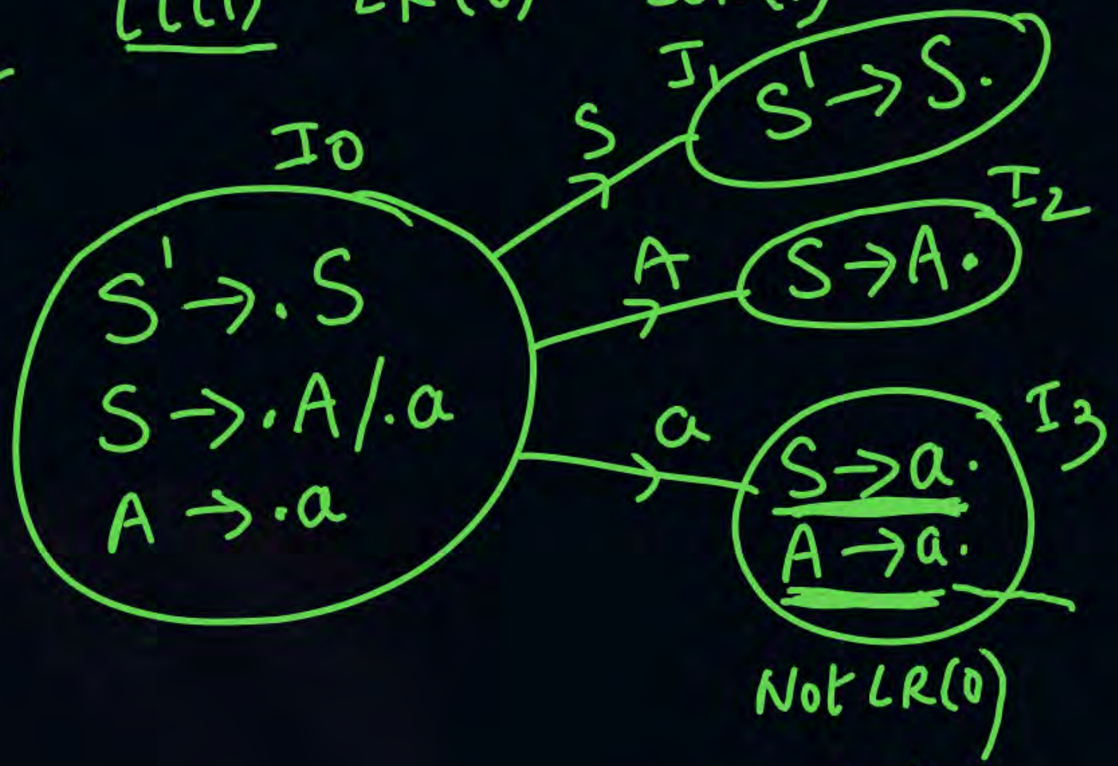
LL(1) ✓

	a
S	$S \to A / S \to a$

LL(1) LR(0) SLR(1)

$S \to A_1 / a_2$

$A \to a_3$



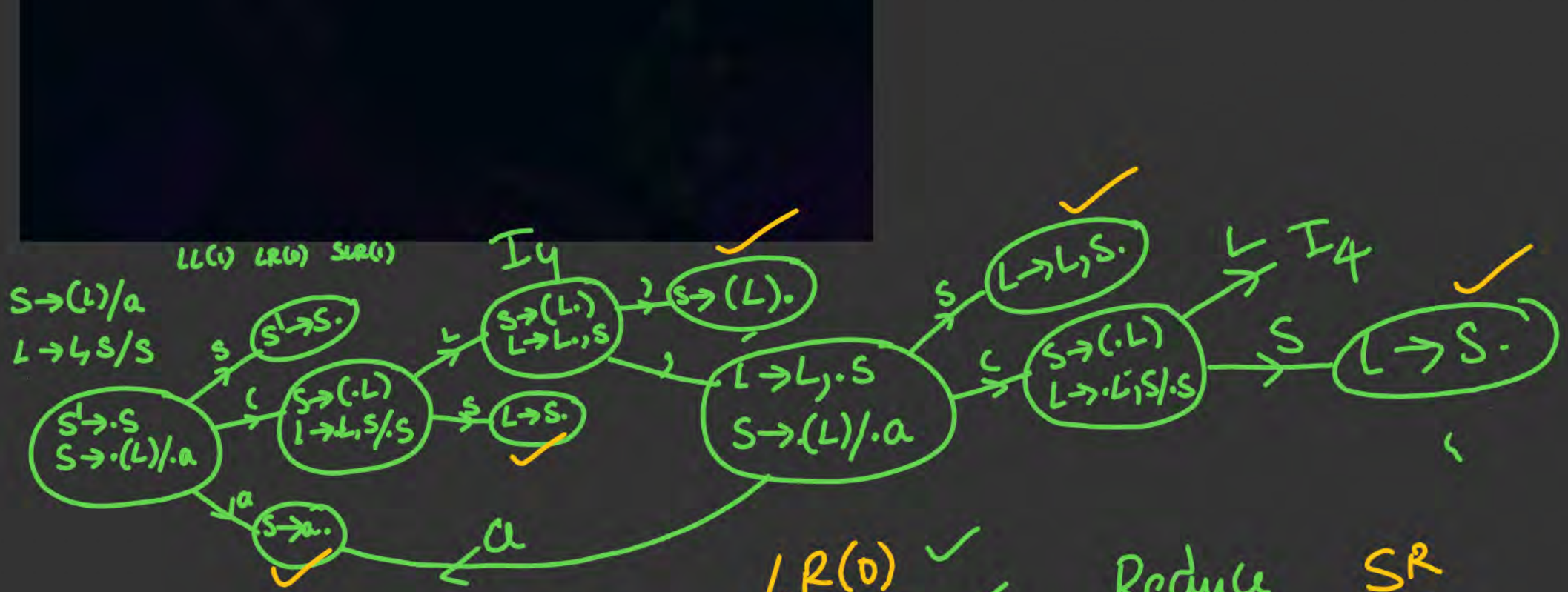
Not LR(0)

	a	$
3		r2 / r3

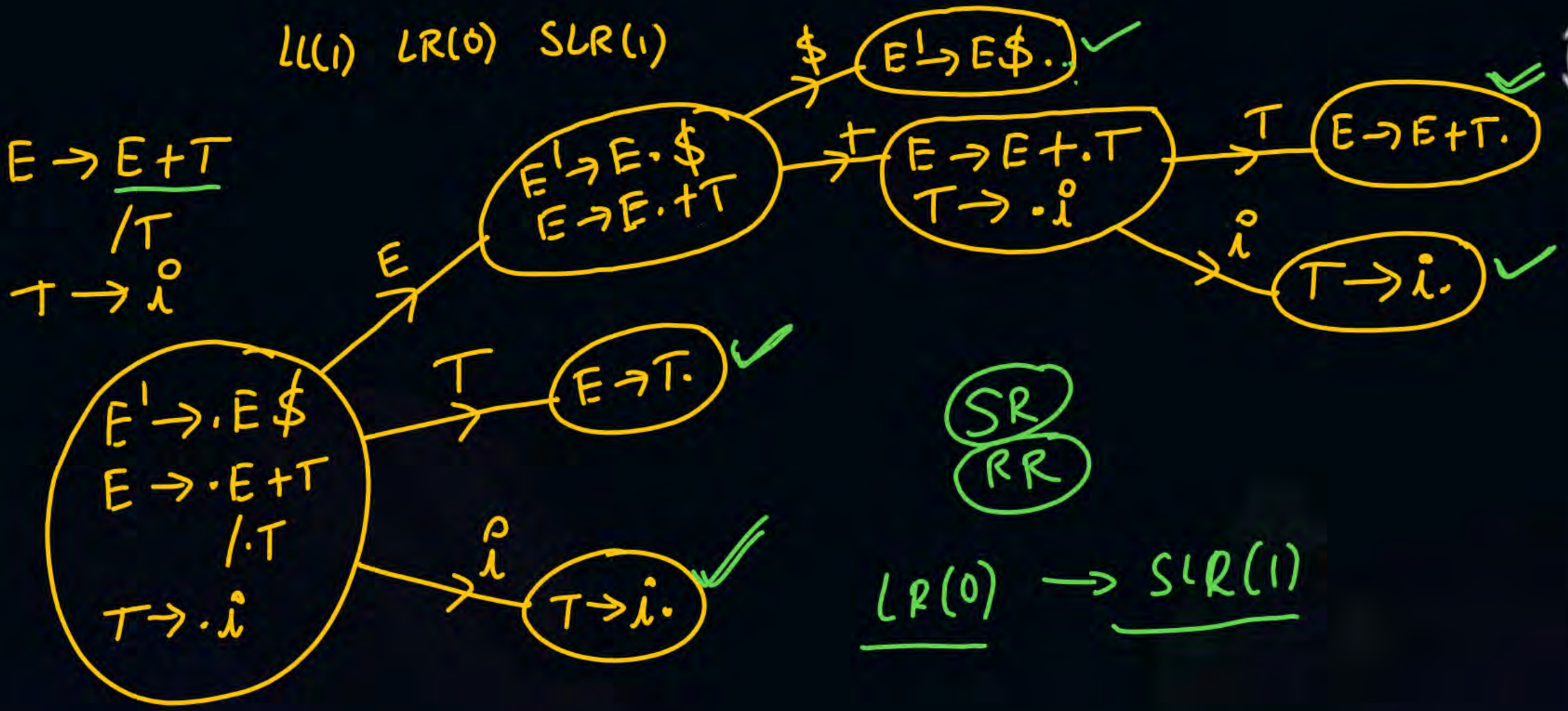
SLR(1) ✗

follow(S) = {$}

follow(A) = {$}


LL(1) LR(0) SLR(1)
S→(L)/a
L→L,S/S
S'→.S
S→.(L)/.a
S'→S.
S→(.L)
L→L,S/.S
L→S.
S→a.
S→(L.)
L→L,S
S→(L).
L→L,.S
S→(L)/.a
L→L,S.
S→(.L)
L→.L,S/.S
L→S.
I4
I4
LL(1) X
LR(0) ✓
SLR(1) ✓
Reduce
SR
RR
L→L,S/S
Any left recursive grammar is not LL(1)

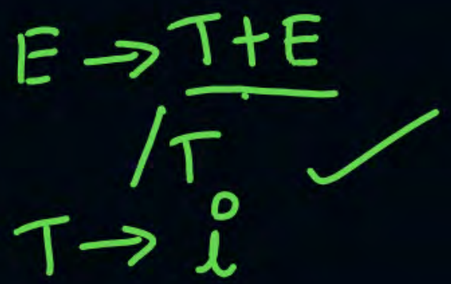
LL(1) LR(0) SLR(1)



SR

RR

LR(0) → SLR(1)


E → T+E
T
T → i

THANK - YOU